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Toward blind joint demosaicing and denoising of raw color filter array data

Fangfang Wu [a], Tao Huang [a], Weisheng Dong [a,*], Guangming Shi [b], Zhonglong Zheng [c], Xin Li [d]

[a] The State Key Laboratory on Integrated Services Networks (ISN), The School of Artificial Intelligence, Xidian University, Xi'an 710071, China
[b] School of Artificial Intelligence, Xidian University, Xi'an 710071, China
[c] The Department of Computer Science, Zhejiang Normal University, Jinhua 321004, China
[d] Lane Dept. of Computer Science and Electrical Engineering, West Virginia University, Morgantown WV 26506-6109, USA

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ABSTRACT

Raw color-filter-array (CFA) data collected in the real world are often noisy and signal-dependent, which makes it difficult to recover the full-resolution noise-free color image. Denoising and demosaicing are two popular tools developed for noisy CFA data in modern color imaging pipeline. However, most existing works on joint demosaicing and denoising (JDD) are based on ad hoc assumptions about image degradation process; while in practice little is known about noise statistics (e.g., noise level) and processing pipeline (e.g., gamma correction). We advocate a blind formulation of joint demosaicing and denoising (bJDD) problem in this paper and present a novel divide-and-conquer approach toward blind reconstruction from noisy raw CFA data. Instead of making over-simplified assumptions about noise statistics, we propose to develop a more realistic Poisson-Gaussian noise model for simulating noisy raw CFA data in the real world. We also introduce a sub-network to adaptively estimate the noise level map from the noisy input, which will provide supplementary information to the deep model for non-blind JDD. Finally, we have adopted a generative adversarial network (GAN) based network for further perceptual optimization. Our extensive experimental results have shown convincingly improved performance over existing state-of-the-art methods in terms of both subjective and objective quality metrics.

1. Introduction

Joint demosaicing and denoising (JDD) refers to an ill-posed problem of reconstructing a full-resolution color image from its incomplete and noisy observations such as Bayer pattern [1]. Due to its importance to color imaging pipeline, both image demosaicing and image denoising have been extensively studied in the past twenty years. Existing approaches toward JDD can be classified into two broad categories: *model-based* and *learning-based*. Model-based approaches focus on the construction of mathematical models (statistical, PDE-based, sparsity-based) in the spatial-spectral domain facilitating the recovery of missing data. Model-based demosaicing techniques such as [2–4] are built upon well-established mathematical models for image restoration such as Total Least Square (TLS) fv [5], Local Polynomial Approximation (LPA) [6] and total-variation (TV) [7]. A common weakness of model-based approaches is the potential mismatch between over-simplified models and sophisticated real-world data especially in view of hand-crafted model parameters.

Learning-based approaches toward JDD have attracted increasingly more attention in recent years. Early works (e.g., [8,9]) using a simple fully connected network only achieved limited success; later works based on Support Vector Regression [10] or Markov Random Fields [11] were capable of achieving comparable performance to model-based demosaicing. Sparsity-based JDD methods leveraging the local adaptation ability of dictionary learning [12,13] can well exploit the correlations between the color channels, achieving promising performance [14,15]. Most recently, inspired by the successes of deep convolutional neural network (DCNN) for both high-level and low-level vision problems [16] - e.g., image recognition [17,18], face recognition [19], image super-resolution [20] and image denoising [21], DCNN has also been applied for JDD [22–25]. Through end-to-end training, state-of-the-art JDD results can be achieved by leveraging the powerful learning ability of DCNN.

The motivation behind this work is largely twofold. On one hand, most of previous JDD methods (e.g., [22]– [25]) used simplified Gaussian noise model independent from image signals. While these methods can achieved good results for simulated noisy CFA, their performance often degrades rapidly for real-world image

* Corresponding author.

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data. This is because signal-independent Gaussian noise model can not accurately characterize the noise distribution in the real-world images. To address this issue, we propose to use a more realistic noise model (Poisson-Gaussian [26]) to simulate the noisy RAW CFA data for training deep models. We have also developed a noise level estimation sub-network to estimate a noise level map from the input inspired by recent work CBDNet [27]. The estimated noise map will facilitate the solution by reducing the original bJDD problem to a non-blind version.

On the other hand, rapid advances in generative adversarial networks (GAN) [28] have found many successful applications from super-resolution (SRGAN [20], ESRGAN [29]) to style transfer (CycleGAN [30], conditional GAN [31]). Through competing with a discriminator network, the generator network can be trained to deliver more photo-realistic reconstruction as demonstrated in SRGAN [20]. Despite the blind formulation, we have found that high-quality color images can serve as the reference for evaluating the visual quality of reconstructed images. Such observation inspired us to design a specially tailored discriminator network for perceptual optimization of bJDD. Moreover, we suggest that the adoption of Relativistic average GAN (RaGAN) [32] further facilitates the objective of improving subjective quality.

The key contributions of this work are summarized as follows:

1. Unlike existing methods based on over-simplified noise models, we propose to adopt a more realistic noise model to train the developed JDD model. Similar to recent blind image denoising method CBDNet [33], a separated noise-level estimation sub-module was proposed to facilitate the task of blind JDD. Unlike CBDNet [33], we have developed a novel noise-level estimation network working with RAW CFA data instead of full-resolution color images.
2. We propose a GAN-based approach toward JDD in which a discriminator network with both perceptual and adversarial loss functions are used for subjective quality assurance. Our generative network is based on deep residue learning (similar to that of [23]) but with the introduction of a new discriminator network. We show GAN-based JDD is capable of delivering *perceptually* optimized reconstruction results.
3. Our extensive experimental results have shown convincingly superior performance over existing state-of-the-art methods in terms of both subjective and objective quality metrics. Specifically, the improvements over previous methods are even more impressive for signal-dependent noise models and real-world noisy RAW images.

The rest of the paper is organized below. In Section 2, we formulate the problem of JDD and discuss some motivation behind. In Section 3, we present the proposed GAN-based joint demosaicing and denoising approach and elaborate the issues related to network architecture, loss function and end-to-end optimization. In Section 4, we report our experimental results and compare them against several other competing approaches. We draw some conclusions about this research in Section 5.

2. Problem formulation and motivation

As mentioned in [22], demosaicing and denoising are traditionally treated as two separated problems and studied by different communities. In practice, raw CFA data are often contaminated by sensor noise [2], which could lead to undesirable artifacts in reconstructed images. Ad-hoc approaches concatenating these two operations often fail: (1) denoising before demosaicing is difficult due to unknown noise characteristics and the challenging in exploiting the correlations between the color channels; (2) denoising after demosaicing is challenging as well because interpolating noisy CFA has intrinsic limitations and will complicate the separation of noise from signal in the spatial domain. Therefore, joint demosaicing and denoising (JDD) has been conceived a more principled approach in terms of problem formulation. Since both demosaicing and denoising are ill-posed, a common model-based image prior can be introduced to facilitate the solution to JDD; various mathematical models have been developed as image priors - e.g. [34,35,6,36–38].

Data-driven approaches toward JDD also exist in the literature such as [39,40,22–25]. Among them, [22] represents an early attempt in which a deep neural network is trained using a large corpus of images. More recent advances include deep residue learning [23], regularization-inspired design of residue denoising networks [24,25]. Despite those progress, we argue that there are two fundamental issues that have been largely overlooked. First, most previous works assumed a signal-independent Gaussian noise model. Though promising JDD results can be achieved on synthetic datasets, the actual performance decreased significantly when applied to real-world noisy CFA image due to the mismatch between the Gaussian noise model and the real-world noise characteristics. Second, previous works opted to train the network by minimizing the mean squared error (MSE) of reconstruction, which is equivalent to the maximization of peak-signal-to-noise ratio (PSNR). However, it is well known that PSNR does not always faithfull correlate with perceptual quality of reconstructed images (e.g., [20]). For practical problems such as blind JDD (no groundtruth is available), it is often more desirable to pursue the solution capable of perceptual instead of PSNR optimization.

2.1. Why signal-dependent noise model?

In the past decades, spatial resolution of CCD/CMOS imaging sensors have increased significantly (e.g., by 1–2 orders of magnitude). As the pixel size becomes smaller and smaller, the signal-to-noise ratio (SNR) of the captured image typically decreases as the fundamental laws in physics dictate, especially for low-light conditions such as night-time environments. It has been shown in [26] that the noise introduced into the RAW CFA-data output of the sensor mainly consists of two components: *signal-independent* noise and *signal-dependent* noise. The signal-independent component mainly arises from electrical and thermal noise [41]; while the signal-dependent counterpart comes from the photon noise observing a Poisson distribution [42]. Since the additive Gaussian noise model widely used in previous works [22,23] can only model the signal-independent component, their performance on real-world RAW CFA data often remains far from being optimized.

In [26], a Poisson-Gaussian noise model was proposed to model the noise of RAW images, where the signal-dependent photon noise was modeled by the Poisson model and the signal-independent electrical/thermal noise was modeled by the Gaussian model. The algorithms for noise model parameters estimation have also been studied in [26,43]. The Poisson-Gaussian noise model has demonstrated its effectiveness in [44,33] for blind image denoising applications. The mixture of Poisson distributions has also been proposed to model the real-world noise in [45], showing promising denoising performance. However, those existing works still suffer from two limitations. First, they are developed for full-resolution noisy color images and cannot be directly applied to noisy CFA data. Second, the noise parameters have to be calibrated for different camera settings. We will address these issues by considering a more realistic noise model in in Section 3.1 and developing a noise level estimation network (NLENet) in Section 3.2.

2.2. Why Generative Adversarial Networks (GAN)?

The basic idea of GAN is to formulate a minimax two-player game by concatenating two competing networks (a generative and a discriminative). In the original setting, the generative model G captures the data distribution and the discriminative model D estimates whether a sample is from the model distribution or the data distribution (real vs. fake). Later GAN was successfully leveraged to the application of image super-resolution (SR) producing photo-realistic images [20], which has inspired a flurry of GAN-based image processing applications including image synthesis (e.g., styleGAN [46] and bigGAN [47]) and style transfer (e.g., CycleGAN [30], conditional GAN [31]). In the setting of JDD, the goal of the generator is to fool the discriminator by generating perceptually convincing reconstructed color images that can not be distinguished from the real ones; while the goal of discriminator is to tell apart the real ground-truth images from those reconstructed by the generator. Through the competition between generator and discriminator, we can pursue a closed-form optimization of JDD as follows:

$$\underset{G}{min}\,\underset{D}{max}V(D_{\Theta_D}, G_{\Theta_G}) = E_{x \sim P_{data}(x)}[logD_{\Theta_D}(x)] + E_{z \sim P_z(z)}[log(1 - D_{\Theta_D}(G_{\Theta_G}(z)))]. \quad (1)$$

where P_{data} is the data distribution (real), P_z is the model distribution (fake), z is the input, x is the label (real or fake).

GAN has received increasingly more attention in recent years. Despite its capability of generating images of good perceptual quality, GAN is also known for its weakness such as difficulty of training (e.g., mode collapse, vanishing gradients etc.), which often results in undesirable artifacts in the reconstructed images. To overcome this difficulty, a set of constraints on network topology was proposed in [48] to address the issue of instability; a conditional version of GAN was constructed in [49] by simply feeding the labeled data. In [50], an energy-based Generative Adversarial NetworkGAN (EBGAN) views the discriminator as an energy function and exhibits more stable behavior than regular GANs. In [51], the Earth-Mover (EM) distance or Wasserstein distance was introduced to GAN which can effectively improve the stability of learning; this idea was further extended in [52] which proposed an alternative to clipping weights of Wasserstein GAN (WGAN). Most recently, Relativistic average GAN (RaGAN) [32] was shown to outperform standard GAN in terms of both speed and performance.

In the scenario of JDD, we note that all existing deep learning based approaches [22–25] have not considered the adoption of GAN because they solely use PSNR as the quality metric of reconstructed images. In view of the limitations with PSNR-based evaluation, we argue a more plausible approach is to consider subjective instead of objective optimization. The class of GANs allow us to feed the perceptual difference (produced by the discriminator) back to the generator, which forms a closed-loop optimization. When compared against previous deep learning-based approach toward JDD (e.g., [22]), we believe that our GAN-based end-to-end optimization has the advantage of learning the bJDD process in a supervised manner and therefore has the potential of better matching with the real-world RAW CFA data. The issue of perceptual optimization and the choice of loss functions will be elaborated in Section 3.3.

3. Proposed method

Blind joint demosaicing and denoising (bJDD) can be formulated as an ill-posed inverse problem in which the forward degradation process is characterized by:

$$\boldsymbol{y} = \mathbf{F} \odot \boldsymbol{x} + \boldsymbol{n}. \quad (2)$$

where $\boldsymbol{x}$ is the original full-resolution color image, $\mathbf{F}$ is 3-dimensional binary matrix indicating missing values in Bayer pattern, $\odot$ denotes element-wise multiplication, $\boldsymbol{n}$ is the vector representing additive noise and $\boldsymbol{y}$ is noisy CFA observation. Then JDD refers to the problem of estimating unknown $\boldsymbol{x}$ from noisy and incomplete observation $\boldsymbol{y}$; when little is known about the noise type and statistics in $\boldsymbol{n}$, we deal with blind JDD. Due to its ill-posed nature, one has to incorporate a priori knowledge (often called regularization) about both signal $\boldsymbol{x}$ and noise $\boldsymbol{n}$ into the solution algorithm.

Inspired by the success of deep learning in image restoration applications, we propose to tackle the problem of blind JDD by learning a nonlinear mapping from the space of degraded image $\boldsymbol{y}$ to the clean full-resolution color image $\boldsymbol{x}$ as shown in Fig. 1. Specifically, our generator network consists of two subnetworks: noise level estimation (NLE) network and JDD network. The former is responsible for estimating a spatially and channel adaptive map characterizing the noise level in RAW CFA data; the latter builds upon the estimated noise level map and solves a non-blind JDD problem (conceptually similar to recent work CBDNet [27]). In the following sections, we will first introduce the Poisson-Gaussian noise model for simulating the noisy CFA images, and then describe the implementation details of the network architecture, loss function and training procedure.

3.1. The Poisson-Gaussian noise model

It has been shown in [26] that the noise contained in the RAW CFA data of common CCD sensor output mainly consists of two parts - i.e., the signal-dependent and independent components. Instead of using a signal-independent Gaussian model, the Poisson-Gaussian noise model developed in [26] is adopted to more accurately characterize the noise in the real-world CFA RAW data. Poisson-Gaussian noise model consists of the signal-dependent component is modeled by the Poisson model and the signal-independent component is modeled by the Gaussian model. Formally, the Poisson-Gaussian noise model can be expressed as $W \sim \mathcal{N}(0, \sigma^2)$ where

$$\boldsymbol{\sigma}^2 = a\boldsymbol{x} + b. \quad (3)$$

The variance term consists of two components: $a\boldsymbol{x}$ ($\boldsymbol{x}$ are the noiseless pixel values) are spatially varying variances of signal-dependent photon noise and b is the variance of the signal-independent Gaussian noise.

Based on the above Poisson-Gaussian model, we continue to simulate noisy CFA RAW data as follows: $\boldsymbol{y} = \min(\max(\boldsymbol{x} + \mathcal{N}(0, \boldsymbol{\sigma}^2), 0), 1)$, where the noise level depends on the pixel value. As shown in [53,33], estimating the noise levels and taking the noise levels as inputs into the network is helpful in achieving better denoising performance, especially for blind denoising. In [26], a method was proposed to estimate the parameters a and b from the noisy RAW image based on the piecewise smooth assumption of natural images in the wavelet domain. Since the piecewise smooth assumption may not be valid for complex natural images, we propose to estimate the per-pixel noise variance values by a devoted network (NLENet). In [33] the CNN has been used for noise variance estimation for blind color image denoising. Here we adopt a similar method but adapt it to estimate the noise level map for CFA RAW data by taking down-sampling operations in three color channels into account.

3.2. Network architecture

The proposed deep generator convolutional network architecture is shown in Fig. 1. It contains two sub-networks: the noise level estimation network (NLENet) and the JDD network (JDDNet). Similar to [22], we first rearrange RAW CFA data into a quarter-resolution four-channel image array before feeding into NLENet and JDDNet. The NLENet takes four-channel image as the input and estimates the noise variance for each pixel, producing a four-

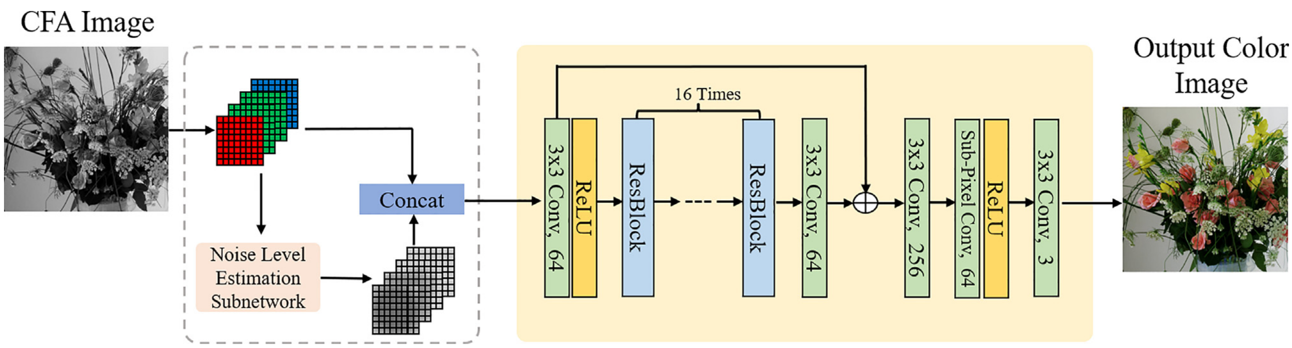

Fig. 1. The overall network architecture of our generator for blind joint demosaicing and denoise.

channel noise level map as the output. The rearranged multi-channel CFA image is then concatenated with the estimated noise variance map to generate a 8-channel image (4-channel RAW CFA data and 4-channel noise level map), which is fed into the JDDNet to produce the final full-resolution color image.

As shown in Fig. 2 (b), the NLENet consists of five convolution (Conv) layers. Each Conv layer except the last layer generates 32-channel feature maps with 3×3 kernels, followed by the ReLU nonlinearity function. The last Conv layer generates the 4-channel variance map. For the JDDNet, the concatenated CFA image and the noise level maps are first fed into a Conv layer with ReLU nonlinearity function, followed by 16 residual blocks (ResBlocks) and a Conv layer to extract the features from the input CFA image. Each ResBlock consists of 2 Conv layers with a ReLU nonlinearity function with a skip connection. To alleviate the notorious gradient vanishing problem [54], a skip connection between the outputs of the first two Conv layers is added, as shown in Fig. 1. A sub-pixel layer proposed in [55] for image super-resolution was adopted to up-sample the CFA image feature maps into the full-resolution color feature maps. All Conv layers use small 3×3kernels to generate 64 feature maps except the last one which has 64×4 feature maps. Finally, in order to restore the full-resolution color image which has three channels, we use a convolution layers with 3×3 kernels and 3 feature maps.

To facilitate the task of subjective quality evaluation, we have designed a separated discriminator network as shown in Fig. 2 (a). Following the architecture proposed in [48], we propose to use a Conv layer followed by batch normalization and LReLU function ($\alpha = 0.2$) as the basic unit throughout the discriminator network. The designed discriminator network is jointly trained with the generator network to solve the two-player minmax problem as described in Eq. (1). It first contains 8 Conv layers with 3×3 kernels, an increasing number of feature size by a factor of 2 from 64 to 512 and stride 2 is used by interval to reduce the resolution. Finally, a Conv layer is used with 3×3 kennel and feature size 1 followed by a sigmoid activation function to gain a probability of similarity score normalized to [0,1]. After setting up the backbone of GAN, we proceed with the discussion of loss functions next.

3.3. Loss functions

Given a training set $\{\boldsymbol{y}_i, \boldsymbol{x}_i\}_{i=1}^{N}$, where $\boldsymbol{y}_i \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times 4}$ is a rearranged 4-channel noisy CFA images and $\boldsymbol{x}_i \in \mathbb{R}^{H \times W \times 3}$ is the original full-color image, we optimize parameters of the networks by minimizing the following loss function

$$\left(\widehat{\Theta}_G, \widehat{\Theta}_E\right) = \mathrm{argmin}_{\Theta_G, \Theta_E} \frac{1}{N} \sum_{i=1}^{N} \left\{ L_{rec}\left(\left(G_{\Theta_G}(\boldsymbol{y}_i), \boldsymbol{x}_i \right) + \lambda_n \left(E_{\Theta_E}(\boldsymbol{y}_i) - \sigma_{i,n} \right)^2 \right\}, \tag{4}$$

where Θ_G and Θ_E denote the parameters of the JDDNet and NLENet, respectively, $L_{rec}(\cdot)$ denotes the image reconstruction loss, λ_n is the parameter introduced to control the tradeoff among the two losses, and $\sigma_{i,n}$ is the ground truth noise variance maps of the i-th training image $\boldsymbol{y}_i$. Here, for simplicity we have used the mean square error (MSE) loss for the noise variance estimation. Regarding the image reconstruction loss, in addition to the MSE loss two ways are considered for further improving the image reconstruction quality: (1) to introduce a perceptual loss depending on high-level features for better characterizing the subjective quality of an image (often requiring a pre-trained network); (2) to introduce a discriminator network whose objective is to learn to distinguish the difference between real and reconstructed (fake) images. In this paper, we propose to combine both ideas and formulate the following composite loss function for the blind JDD problem:

$$L_{rec} = L_{MSE} + \lambda_p L_p + \lambda_A L_A. \tag{5}$$

where L_{MSE} is the conventional per-pixel loss function such as mean square error, L_p is the perceptual loss given by a pre-trained loss network and L_A is the adversarial loss associated with the discriminator of GAN. Two Lagrangian parameters (λ_p and λ_A) are introduced to control the tradeoff among those three regularization

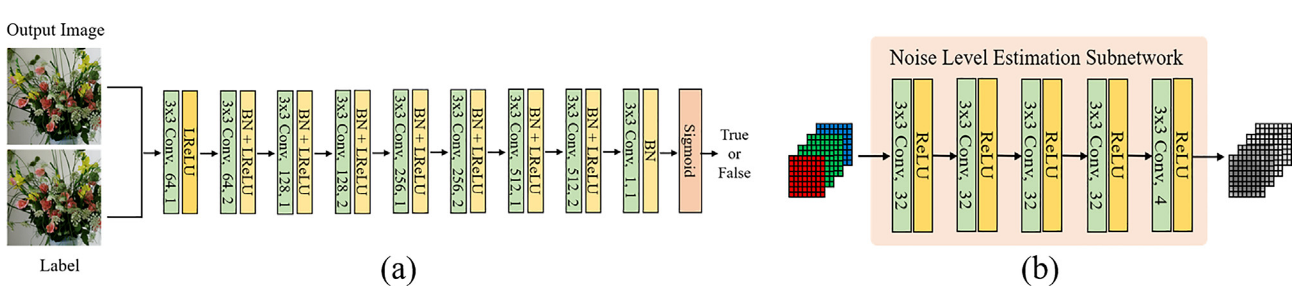

Fig. 2. The network architectures of (a) the discriminator and (b) the noise level estimation subnetworks (NLENet).

terms. Detailed formulation and implementation of these three loss functions are provided as follows.

MSE loss. Given an image pair $(\boldsymbol{y}_i, \boldsymbol{x}_i)$, the MSE loss is given by:

$$L_{MSE} = \frac{1}{3HW}||G_{\Theta_G}(\boldsymbol{y}_i) - \boldsymbol{x}_i||_2^2. \tag{6}$$

Perceptual loss. We adopted recently proposed Perceptual loss [56]. It is a simple $L2$ metric defined as the loss based on the ReLU activation layers of the pre-trained 19 layer visual geometry group (VGG) network described in [17]. The perceptual loss term is given by:

$$L_p = \frac{1}{C_f H_f W_f}\sum_{j=1}^{3}||\phi_j(\boldsymbol{x}_i) - \phi_j(G_{\Theta_G}(\boldsymbol{y}_i))||_2^2, \tag{7}$$

where $\phi_j(\boldsymbol{x}_i)$ denotes the operator extracting the feature maps of $\boldsymbol{x}_i$ of a pre-trained VGG19 network. Here, the feature maps obtained by the $ReLU_{5,4}$ layer of the VGG19 are selected. $C_f \times H_f \times W_f$ denotes the size of the feature maps.

Adversarial loss Instead of using the discriminator, the Relativistic average Discriminator (RaD) of [32] is used for improved performance. Given a set of N reconstructed images generated from the generator $\hat{\boldsymbol{x}}_i = G_{\Theta_G}(\boldsymbol{y}_i), i = 1, \cdots, N$, the RaD is formulated as

$$D_{Ra}(\boldsymbol{x}_i, \hat{\boldsymbol{x}}_i) = sigmoid\left(C(\boldsymbol{x}_i) - \frac{1}{N}\sum_{i=1}^{N} C(\hat{\boldsymbol{x}}_i)\right), \tag{8}$$

where $C(\cdot)$ denotes the non-transformed discriminator. The RaD calculates the averaged probability that the real $\boldsymbol{x}_i$ is more realistic than its fake counterpart $\hat{\boldsymbol{x}}_i$. With the RaD, the adversarial loss for guiding the generator can be defined as:

$$L_A = -\frac{1}{N}\sum_{i=1}^{N}\left[log(D_{Ra}(\hat{\boldsymbol{x}}_i, \boldsymbol{x}_i)] - \frac{1}{N}\sum_{i=1}^{N}[log(1 - D_{Ra}(\boldsymbol{x}_i, \hat{\boldsymbol{x}}_i))]\right. . \tag{9}$$

3.4. Training details

We implemented the networks using TensorFlow platform and the training was performed on a single NVIDIA 1080Ti GPU using a collection of high-quality full-color images. These training images are separated from the testing images and were cropped to patches of size 96×96 (so the size of input data to the generator network is $48 \times 48 \times 4$). Since the models are based on a full convolutional network and are trained on image patches, we can apply the trained model to test images of arbitrary size. We normalize the input and output of networks to [0, 1], so the loss is calculated based on the scale of [0, 1]. In all experiments, we set the weights $\lambda_n = 0.5, \lambda_p = 1$ and $\lambda_A = 0.001$. During the training of the proposed network, we alternately perform gradient descent steps between the generator $(G_{\Theta_G}, E_{\Theta_E})$ and the discriminator D_{Θ_D} using Adam algorithm [57] with parameter $\beta_1 = 0.9$. The learning rate is set to 10^{-4} for both generator and discriminator networks. The batch size is set to 64, which has shown relatively stable convergence process. The whole training process took around 6 days on the machine.

3.4.1. Training data

It is well known that deep learning benefits from a large number of training samples. In order to achieve better performance on the proposed GAN, we have manually selected more than 1400 high-quality color images, from which patches of size 96×96 were extracted as the ground truth. In order to obtain more training data, we adopted a strategy of data augmentation by flipping the patch left-to-right, upside-down and along the diagonal. This way the total amount of training data is increased by a factor of $8\times$, which leads to $320,000$ training pairs for training. To simulate the signal-dependent noise, we use Eq. 3 to generate the noisy Bayer pattern, where the parameters a and b are uniformly sampled from the ranges $[0, 0.26]$ and $[0, 0.004]$, respectively. To compare with other JDD methods that used plain Gaussian noise model, we also generate noisy Bayer pattern with conventional Gaussian noise model with noise variances in the range of $[0, 0.078]$.

3.4.2. Testing data

The McMaster and Kodak datasets are the most common test sets for image demosaicing. The Kodak datasets contains 24 images of size 512×768 derived from scanning of early film-based data sources. Despite the popularity of Kodak data set in image demosaicing community, most Kodak images contain relatively smooth edges and textures whose visual quality are not among the best based on the modern day's criterion. By contrast, the McMaster data set contains 18 images of size 500×500 containing abundant strong and sharp image structures. Most recently, a new data set called Waterloo Exploration Database (WED) [58] with 4,744 high-quality natural images has been made publicly available and adopted by a recent demosaicing study [23]. The MSR dataset [39] containing 200 noise-free and noisy raw image pairs and their corresponding RGB ground truths is also adopted.

4. Experimental results

In this section, we report experimental results with the proposed method. To verify the performance of the proposed method, we first report the experimental results based on Gaussian noise model with comparisons with recently state-of-the-art JDD methods. Then, we present the experimental results with signal-dependent noise model. The following objective quality measures are used to evaluate the performance of different competing methods: Red channel Peak Signal to Noise Ratio (RPSNR), Green channel Peak Signal to Noise Ratio (GPSNR), Blue channel Peak Signal to Noise Ratio (BPSNR), Color Peak Signal to Noise Ratio (CPSNR) and Structural Similarity Index (SSIM).

4.1. Experimental results on uniform Gaussian noise

For conventional Gaussian noise model, we have compared the proposed JDD method with several state-of-the-art JDD methods including Sequential Energy Minimization(SEM) [40], a flexible camera image processing framework (FlexISP) [38], a deep learning method(DJ) [22] and a variant of ADMM [59], using their source code on the same dataset. We also report the experimental results of the proposed generator network without discriminator. Unlike those benchmark methods requiring the prior knowledge about the noise level of Gaussian noise, our method is blind denoise and demosaicing (no such prior is needed). The average PSNR/CPSNR and SSIM results on Kodak and McMaster for noise levels $\sigma_n = 0, 5/255, 10/255, 20/255$ are shown in Table 1. The OURS[1] in the table represents the result of our generator network without the discriminator and the OURS[2] in the table represents the result of our GAN network (with the discriminator). Note that the results for noise level being zero means the JDD problem degenerates to the original demosaicing problem. Similar to [23], we have compared the PSNR result of each color channel in the table. From Table 1, one can see that the proposed method achieves much better average PSNR/CPSNR and SSIM results than all the other competing methods. On the average, our method outperforms the second best method by $4.4dB, 2.3dB, 4.1dB$ and $1.5dB$ on four different noise levels, respectively, verifying the effectiveness and robustness of the proposed JDD method.

Table 1
Average PSNR(dB) and SSIM comparisons for the joint demosaicing and denoise results (the best is in bold).

Method	Kodak					McMaster					WED				
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
								n = 0							
FlexISP [38]	34.30	36.30	34.35	34.98	0.9426	35.18	37.39	33.00	35.19	0.9385	35.03	37.88	34.11	35.68	0.9476
SEM [40]	37.03	40.26	36.45	37.55	0.9691	33.53	36.69	33.16	34.13	0.9301	33.98	37.84	33.77	34.75	0.9534
DJ [22]	33.86	34.78	33.01	33.88	0.9271	32.40	34.52	30.52	32.48	0.8876	32.63	34.82	31.33	32.93	0.9174
ADMM [59]	30.94	33.16	30.80	31.63	0.8873	31.97	34.72	31.30	32.66	0.9052	31.37	33.47	30.62	31.82	0.9075
OURS[1]	**42.37**	45.38	**41.32**	**42.64**	**0.9894**	**39.37**	42.26	37.48	39.17	0.9706	**39.72**	43.40	**39.08**	**40.23**	**0.9827**
OURS[2]	42.20	**45.47**	41.29	42.57	0.9889	39.29	**42.44**	**37.49**	**39.17**	**0.9706**	39.61	**43.70**	39.05	40.23	0.9826
Method								n = 5/255							
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexISP [38]	30.78	32.01	31.15	31.31	0.8694	30.86	32.45	30.23	31.18	0.8627	30.84	32.37	30.05	31.09	0.8701
SEM [40]	34.04	35.60	34.38	34.59	0.9269	31.68	33.89	31.95	32.36	0.8869	32.07	34.48	32.27	32.74	0.9114
DJ [22]	32.93	33.82	32.48	33.08	0.9058	31.86	33.79	30.38	32.01	0.8739	31.99	34.01	31.00	32.33	0.8983
ADMM [59]	31.19	32.41	31.20	31.60	0.8787	32.33	33.90	31.65	32.63	0.8966	31.69	32.69	30.99	31.79	0.8955
OURS[1]	**36.72**	**37.64**	**36.42**	**36.87**	0.9506	**35.97**	**37.23**	34.94	**35.87**	**0.9387**	**35.67**	**37.53**	**35.21**	**35.94**	**0.9539**
OURS[2]	36.66	37.58	36.37	36.82	**0.9506**	35.94	37.18	**34.96**	35.86	0.9386	35.61	37.45	35.19	35.90	0.9538
Method								n = 10/255							
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexIS [38]	28.32	28.81	28.78	28.64	0.7583	28.25	29.13	28.16	28.51	0.7534	28.12	28.92	27.69	28.24	0.7603
SEM [40]	29.22	29.97	30.26	29.78	0.7681	27.90	29.38	28.98	28.68	0.7306	28.20	29.61	29.02	28.87	0.7587
DJ [22]	31.45	32.19	31.32	31.65	0.8731	30.79	32.34	29.72	30.95	0.8467	30.78	32.42	30.05	31.08	0.8668
ADMM [59]	30.72	31.47	30.92	31.04	0.8595	31.57	32.53	31.07	31.72	0.8699	31.02	31.58	30.51	31.04	0.8714
OURS[1]	33.67	**34.30**	**33.70**	**33.87**	0.9125	33.57	34.45	33.08	33.60	0.9092	**33.28**	**34.60**	32.91	**33.47**	0.9260
OURS[2]	**33.68**	34.28	33.67	33.86	**0.9129**	**33.58**	**34.46**	**33.08**	**33.61**	**0.9097**	33.27	34.57	**32.92**	33.46	**0.9266**
Method								n = 20/255							
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexIS [38]	25.13	24.69	25.62	25.15	0.5520	24.95	24.99	25.10	25.01	0.5556	24.88	24.77	24.74	24.80	0.5695
SEM [40]	22.46	23.01	23.65	23.00	0.4425	22.11	23.34	23.74	23.0	0.4415	22.26	23.22	23.45	22.93	0.4605
DJ [22]	28.94	29.49	29.08	29.17	0.8015	28.61	29.73	28.03	28.79	0.7789	28.48	29.67	28.04	28.73	0.8012
ADMM [59]	29.01	29.39	29.37	29.26	0.8132	29.24	29.81	28.88	29.31	0.8046	28.85	29.22	28.52	28.87	0.8197
OURS[1]	30.48	30.98	30.70	30.70	0.8455	30.55	31.32	30.56	30.73	0.8554	30.33	31.33	30.10	30.49	0.8747
OURS[2]	**30.50**	**30.99**	**30.72**	**30.73**	**0.8479**	**30.60**	**31.34**	**30.59**	**30.77**	**0.8572**	**30.35**	**31.33**	**30.13**	**30.52**	**0.8767**

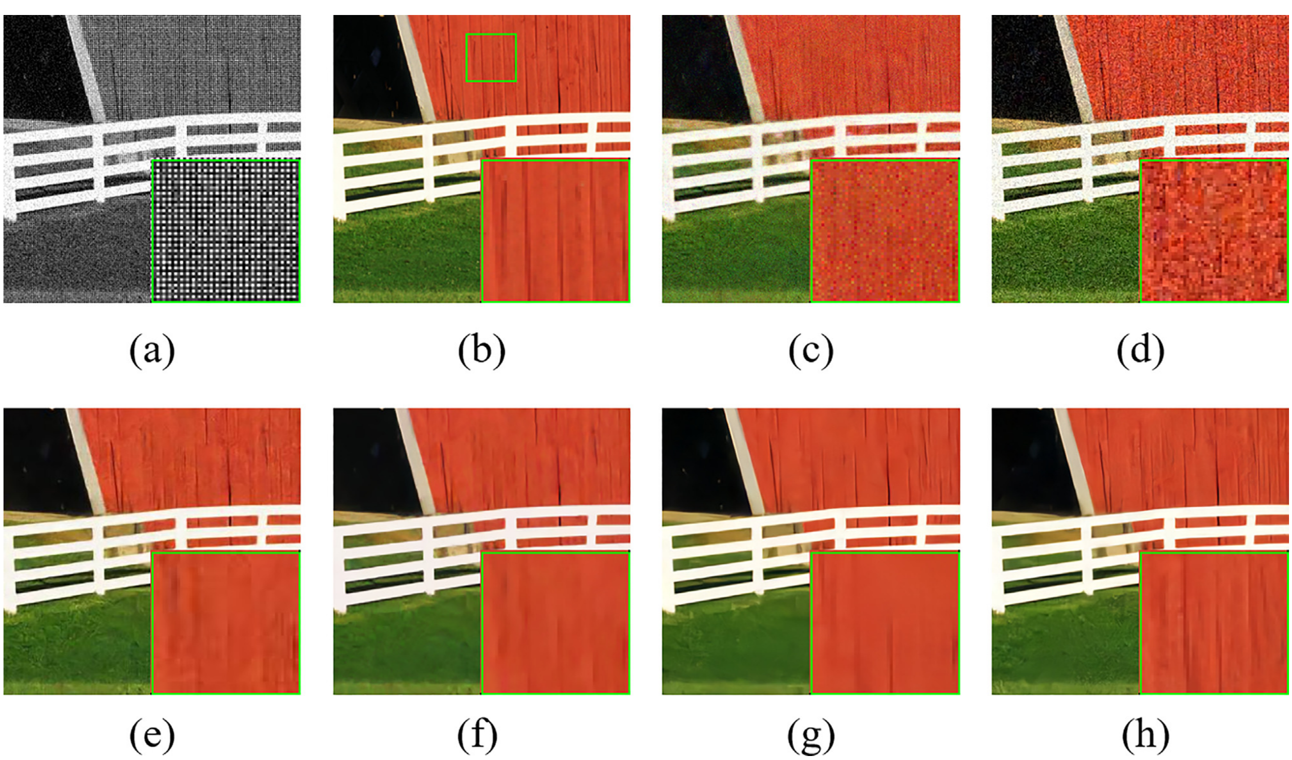

Fig. 3. Visual comparisons on McMaster18 image with uniform Gaussian noise of noise level $\sigma = 20$ for joint denoising and demosaicking. (a) Bayer noisy image; (b) original image; (c) FlexISP [38] (PSNR = 24.80 dB, SSIM = 0.6084); (d) SEM [40] (PSNR = 23.11 dB, SSIM = 0.4183); (e) DJ [22] (PSNR = 28.07 dB, SSIM = 0.7706); (f) ADMM [59] (PSNR = 28.50 dB, SSIM = 0.7862); (g) **Proposed method** without GAN (PSNR = 29.97 dB, SSIM = 0.8308); (h) **Proposed method** with GAN (PSNR = **30.00** dB, SSIM = **0.8387**).

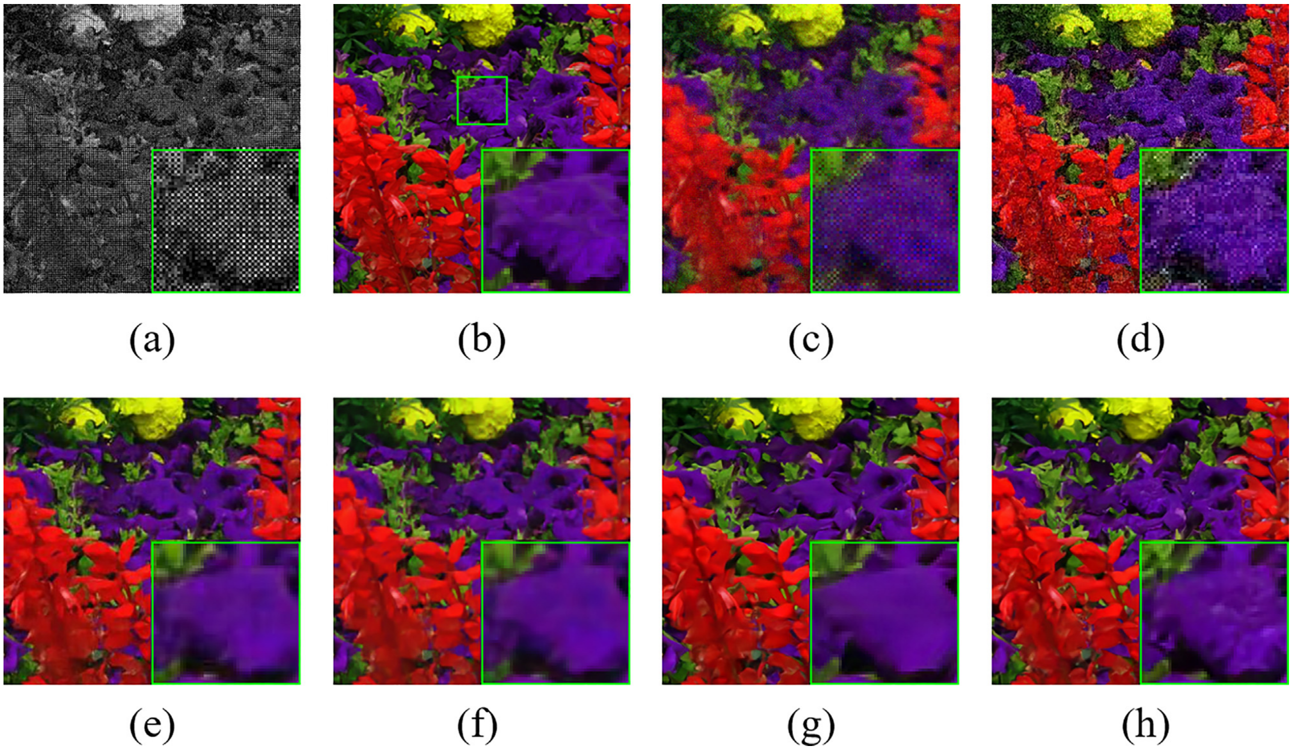

Fig. 4. Visual comparison results of McMaster17 image with uniform Gaussian noise of noise level $\sigma = 20$ for joint denoising and demosaicking. (a) Bayer noisy image; (b) original image; (c) FlexISP [38] (PSNR = 22.80 dB, SSIM = 0.5217); (d) SEM [40] (PSNR = 24.33 dB, SSIM = 0.4072); (e) DJ [22] (PSNR = 26.35 dB, SSIM = 0.7051); (f) ADMM [59] (PSNR = 27.42 dB, SSIM = 0.7586); (g) **Proposed method** without GAN (PSNR = 28.28 dB, SSIM = 0.7966); (h) **Proposed method** with GAN (PSNR = **28.33** dB, SSIM = **0.8000**).

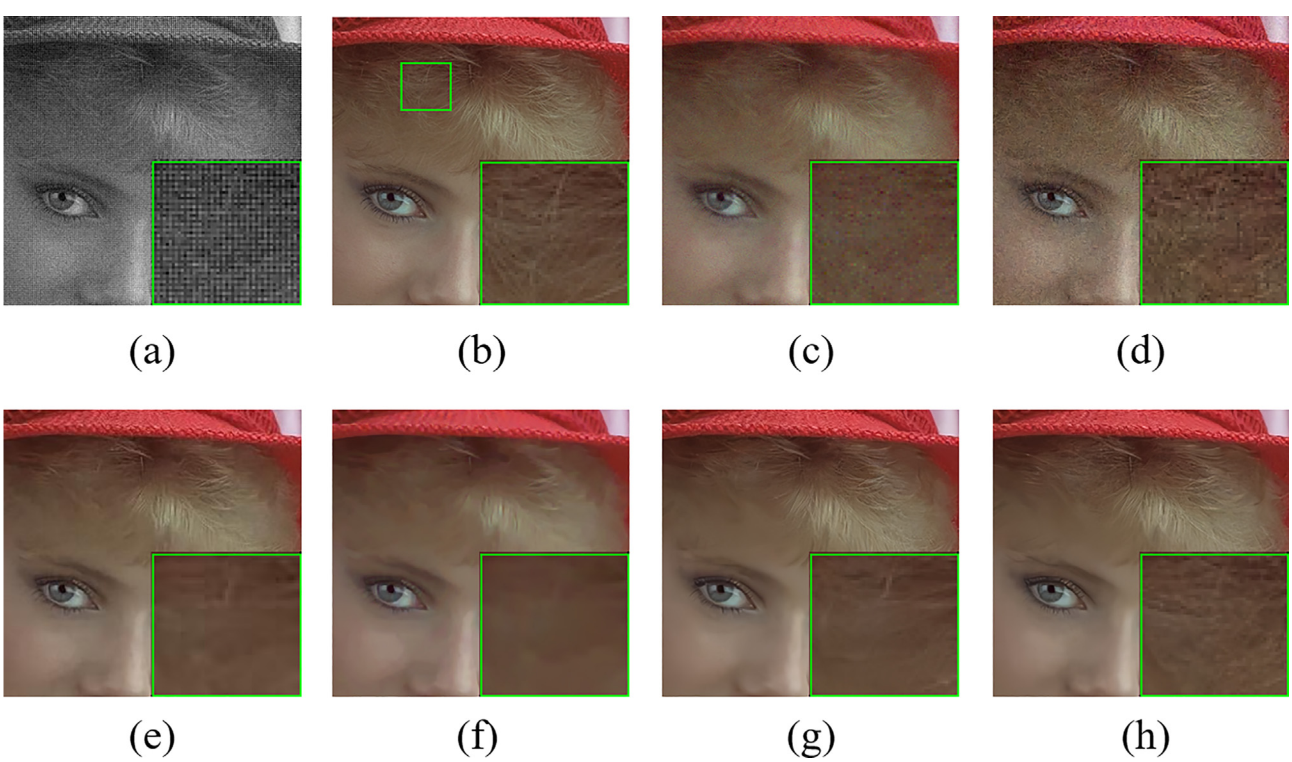

Fig. 5. Visual comparison results of kodak4 image with uniform Gaussian noise of noise level $\sigma = 10$ for Joint denoising and demosaicking. (a) Bayer noisy image; (b) original image; (c) FlexISP [38] (PSNR = 29.67 dB, SSIM = 0.7395); (d) SEM [40] (PSNR = 29.63 dB, SSIM = 0.7055); (e) DJ [22] (PSNR = 32.43 dB, SSIM = 0.8495); (f) ADMM [59] (PSNR = 31.93 dB, SSIM = 0.8414); (g) **Proposed method** without GAN (PSNR = 34.27 dB, SSIM = 0.8912); (h) **Proposed method** with GAN (PSNR = **34.27** dB, SSIM = **0.8928**).

Table 2
The PSNR and SSIM results of the test methods for signal-dependent noise.

Method	McMaster					Kodak				
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexISP [38]	28.82	30.18	29.10	29.11	0.7888	27.36	27.95	27.95	27.60	0.7222
SEM [40]	27.51	29.48	29.08	28.47	0.7695	27.69	28.67	28.90	28.31	0.7386
DJ [22]	29.48	30.66	29.01	29.59	0.8087	29.33	29.86	29.63	29.58	0.7984
ADMM [59]	29.95	30.39	29.99	30.00	0.8364	28.61	28.48	29.15	28.71	0.7718
Proposed method	**34.19**	**35.62**	**34.01**	**34.38**	**0.9297**	**33.93**	**34.88**	**34.07**	**34.22**	**0.9276**
Method	Moiré					VDP				
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexISP [38]	26.34	27.98	26.69	26.67	0.7692	23.71	25.34	24.05	24.07	0.7595
SEM [40]	26.27	28.18	26.99	26.93	0.7745	25.98	27.91	26.43	26.56	0.8473
DJ [22]	27.12	28.44	27.42	27.46	0.7673	25.37	26.56	25.41	25.61	0.7840
ADMM [59]	26.56	27.39	27.13	26.75	0.7381	23.47	24.08	24.10	23.65	0.6486
Proposed method	**30.00**	**32.06**	**30.60**	**30.56**	**0.8790**	**28.99**	**30.65**	**29.34**	**29.43**	**0.9128**

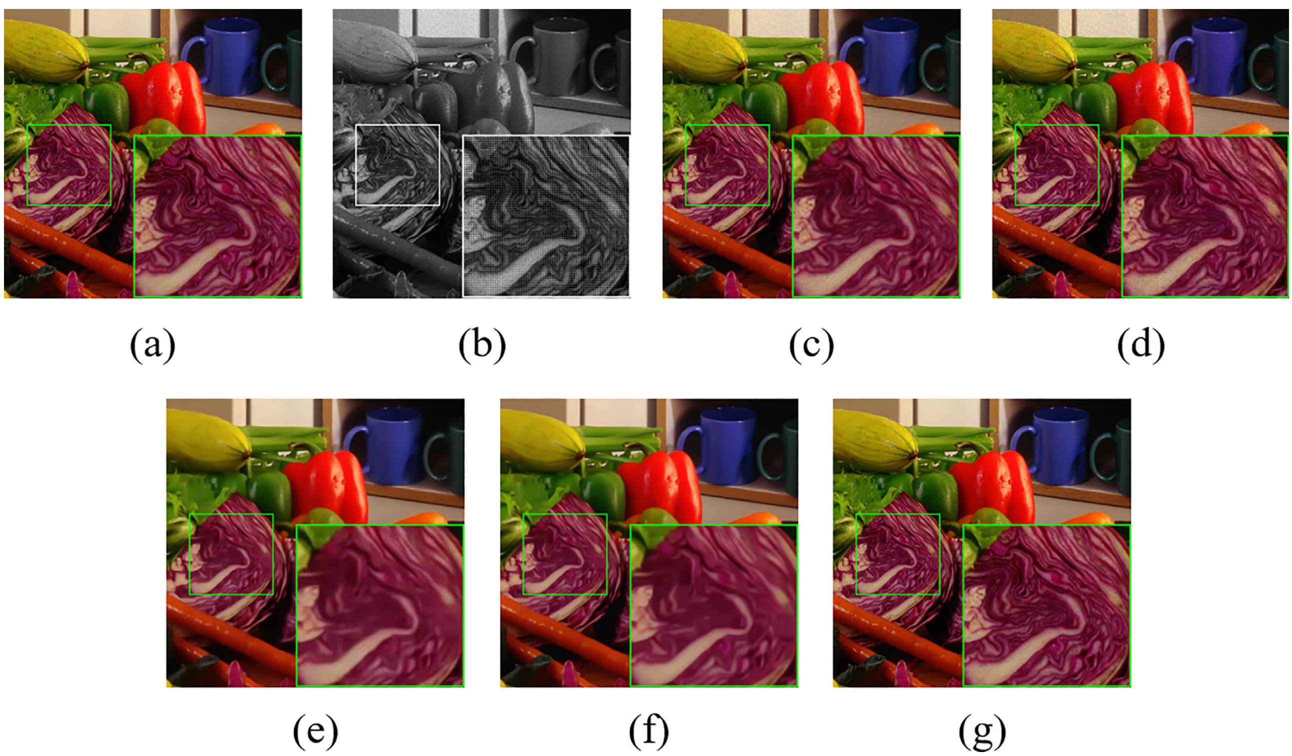

Fig. 6. Visual comparison results on the McMaster dataset for the non-uniform signal-dependent noise. (a) The clean image; (b) the noisy raw image; (c) FlexISP [38] (PSNR = 34.08 dB, SSIM = 0.9086); (d) SEM [40] (PSNR = 33.53 dB, SSIM = 0.9092); (e) DJ [22] (PSNR = 30.48 dB, SSIM = 0.8286); (f) ADMM [59] (PSNR = 33.56 dB, SSIM = 0.8887); (g) **Proposed method** (PSNR = **37.45** dB, SSIM = **0.9553**).

Figs. 3–5 demonstrates the subjective quality comparison results with different noise levels $\sigma_n = 20, 10$. We can observe that the method of SEM suffers from the lack of robustness to noise; the methods of FlexISP and DeepJoint (DJ) both suffer from various artifacts such as vertical color lines, leftover noisy pixels and unnatural color. Among the competing approaches, the ADMM algorithm is relatively good, but when compared with our method still arguably falls behind in terms of visual quality. The reconstructed images by our method visually appear much better in terms of fewer artifacts and more vivid color demonstrating the superiority and robustness of the proposed algorithm. In particular, the proposed method with discriminator can recover more image details than the generator network without discriminator, which justifies the effectiveness of GAN-based perceptual optimization.

4.2. Experimental results on signal-dependent noise

We have also evaluated the performance of the proposed method on images corrupted with signal-dependent noise as generated by Eq. 3. In addition to Kodak and McMaster datasets, we have verified the performance on VDP and Moiré datasets of [22]. For other competing methods that need a noise level as an input parameter, we used the noise estimation method of [43] to estimate the noise variances for each channel and use the averaged variances as the final estimate of the noise variances. Table 2 shows the average results of the competing methods on the four datasets. From Table 2, we can see that the proposed method performs significantly better than the other methods. The proposed method outperforms the ADMM method, which is the second best

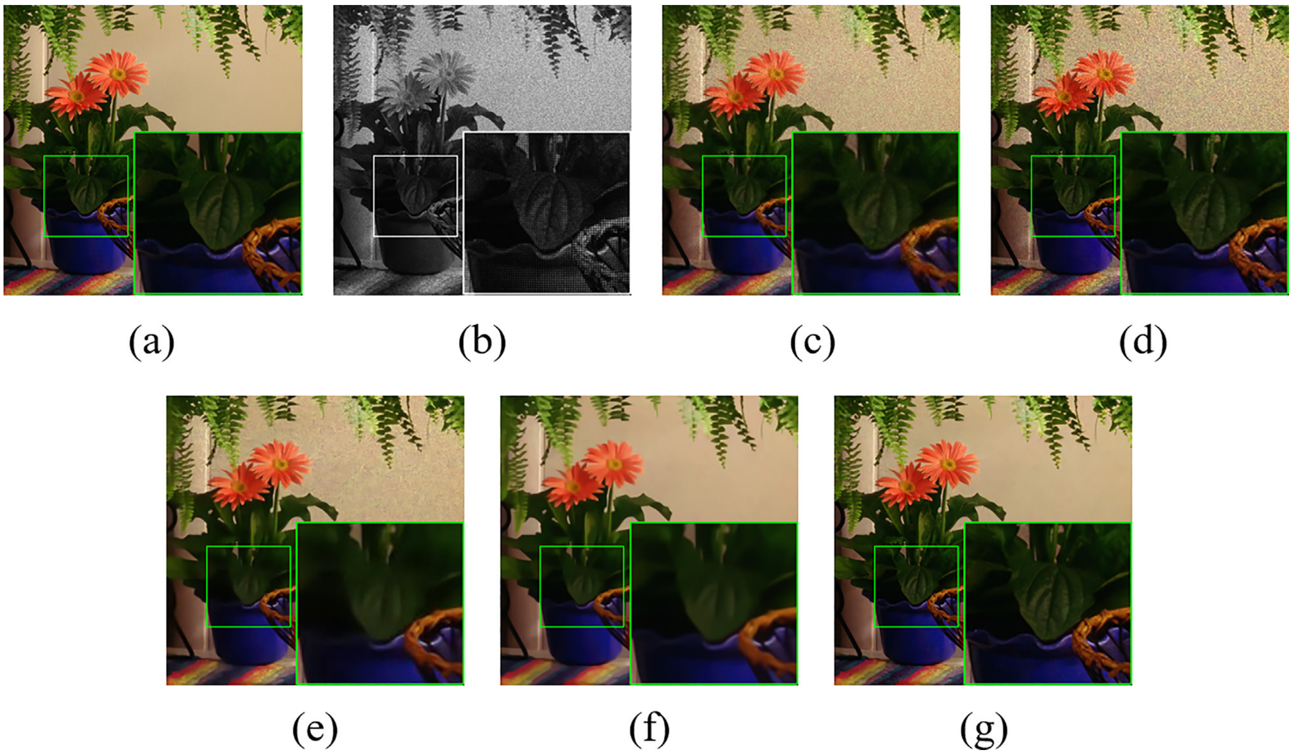

Fig. 7. Visual comparisons results on the McMaster dataset for the non-uniform signal-dependent noise. (a) The clean image; (b) the noisy raw image; (c) FlexISP [38] (PSNR = 25.98 dB, SSIM = 0.6452); (d) SEM [40] (PSNR = 24.86 dB, SSIM = 0.6353); (e) DJ [22] (PSNR = 28.92 dB, SSIM = 0.7614); (f) ADMM [59] (PSNR = 29.55 dB, SSIM = 0.8654); (g) **Proposed method** (PSNR = **33.77** dB, SSIM = **0.9277**).

Table 3
The average PSNR results of the test methods on the MSR dataset [39].

Method	Noise-free		Noisy	
	linRGB	sRGB	linRGB	sRGB
LNRF [39]	39.4	32.6	37.8	31.5
SEM [40]	40.9	34.6	38.8	32.6
DJ [22]	**42.7**	**35.9**	38.6	32.6
CCRDN [24]	41.0	34.6	39.2	33.3
MMNet [25]	42.1	35.6	39.2	33.3
Proposed method	42.1	35.4	**39.9**	**33.8**

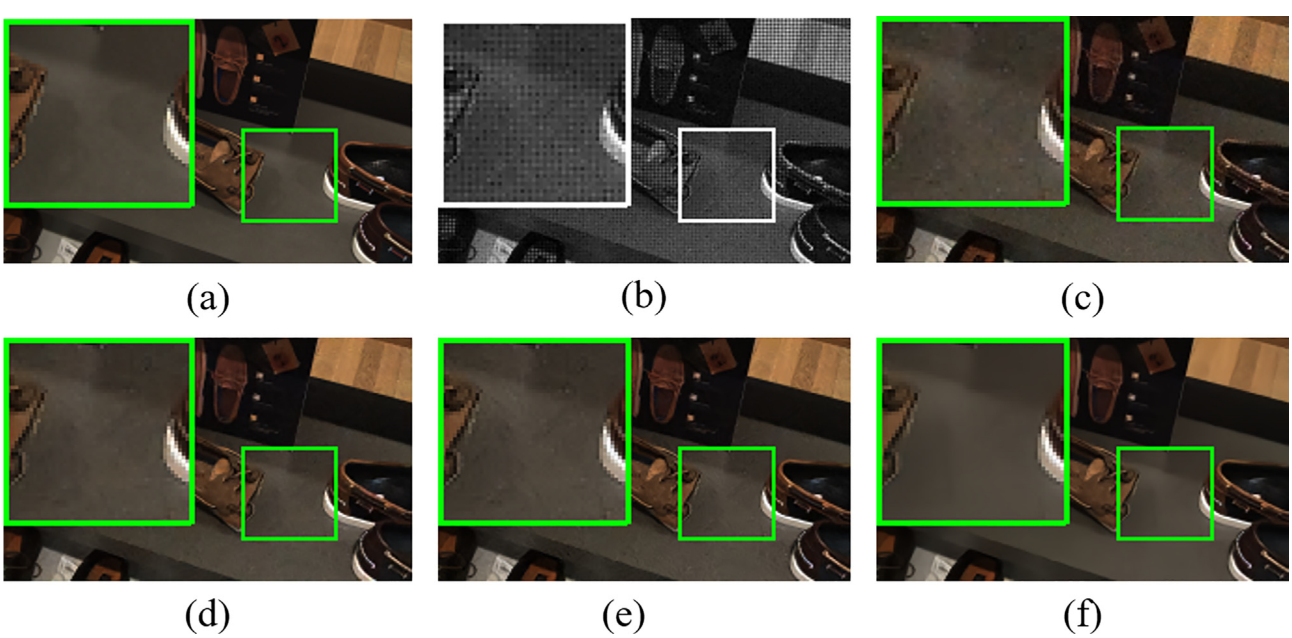

Fig. 8. Visual comparison results on the MSR test dataset [39] for the non-uniform signal-dependent noise. (a) The clean image; (b) the noisy raw image; (c) LNRF [39] (PSNR = 35.05 dB); (d) SEM [40] (PSNR = 36.94 dB); (e) MMNet [25] (PSNR = 37.03 dB); (f) **Proposed method** (PSNR = **38.37** dB).

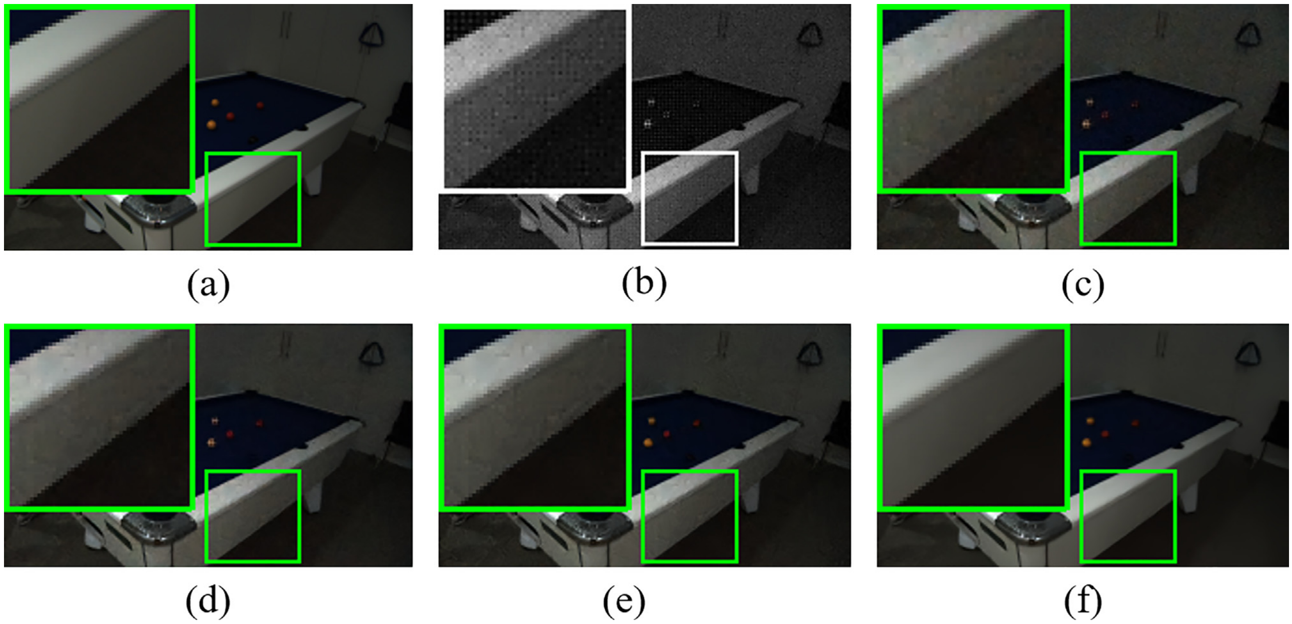

Fig. 9. Visual comparison results on the MSR test dataset [39] for the non-uniform signal-dependent noise. (a) The clean image; (b) the noisy raw image; (c) LNRF [39] (PSNR = 36.16 dB); (d) SEM [40] (PSNR = 37.74 dB); (e) MMNet [25] (PSNR = 38.27 dB); (f) **Proposed method** (PSNR = **40.99** dB).

Table 4
The PSNR and SSIM results of the test methods on real noisy images from SIDD [44].

Method	SIDD [44]				
	RPSNR	GPSNR	BPSNR	CPSNR	SSIM
FlexISP [38]	35.41	38.22	36.15	36.31	0.8780
SEM [40]	36.87	39.53	38.15	37.98	0.9156
DJ [22]	36.68	39.70	37.44	37.67	0.9327
ADMM [59]	37.97	36.49	38.77	37.48	0.9454
Proposed method	**39.44**	**42.24**	**39.53**	**40.10**	**0.9563**

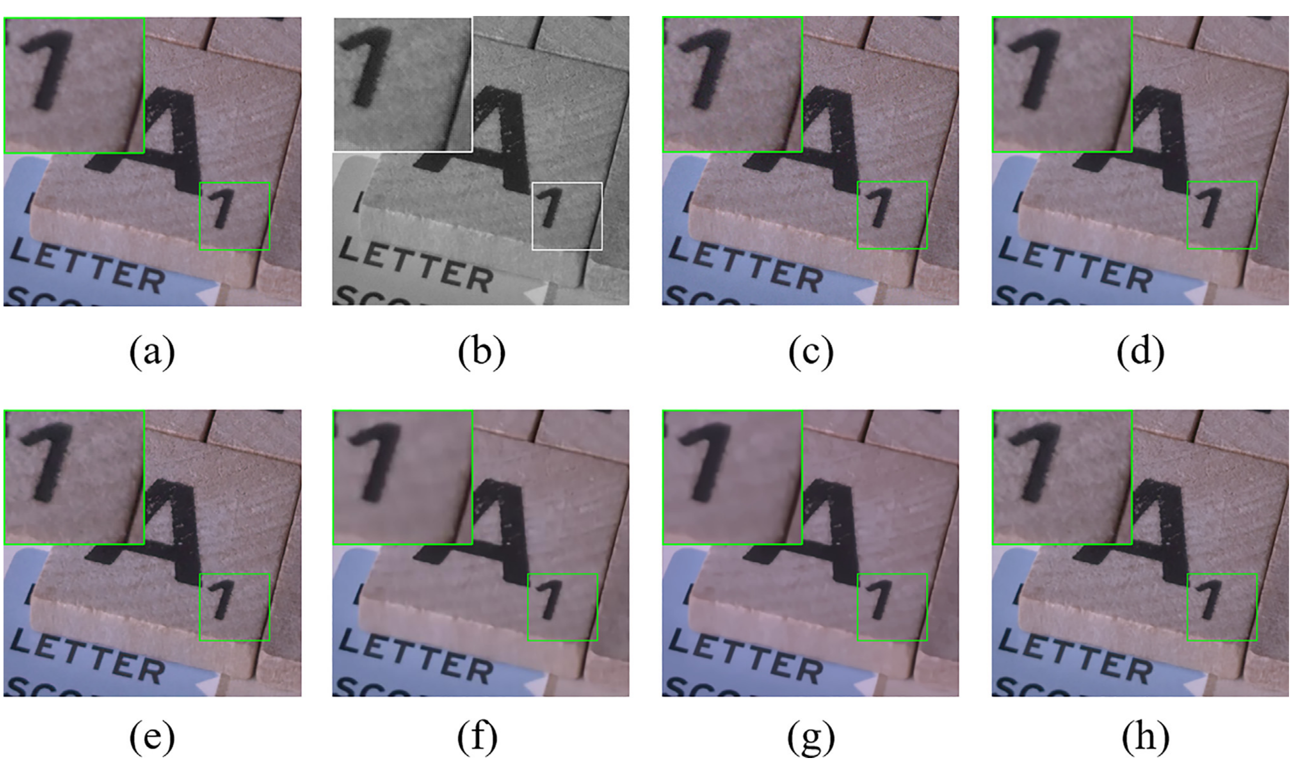

Fig. 10. Visual comparison results for the real noisy images from the SIDD dataset. (a) The clean image; (b) the noisy raw image; (c) the noisy sRGB image (PSNR = 37.63 dB, SSIM = 0.9212); (d) FlexISP [38] (PSNR = 37.67 dB, SSIM = 0.9307); (e) SEM [40] (PSNR = 38.75 dB, SSIM = 0.9555); (f) DJ [22] (PSNR = 36.12 dB, SSIM = 0.8934); (g) ADMM [59] (PSNR = 36.01 dB, SSIM = 0.9086); (h) **Proposed method** (PSNR = **39.97** dB, SSIM = **0.9615**).

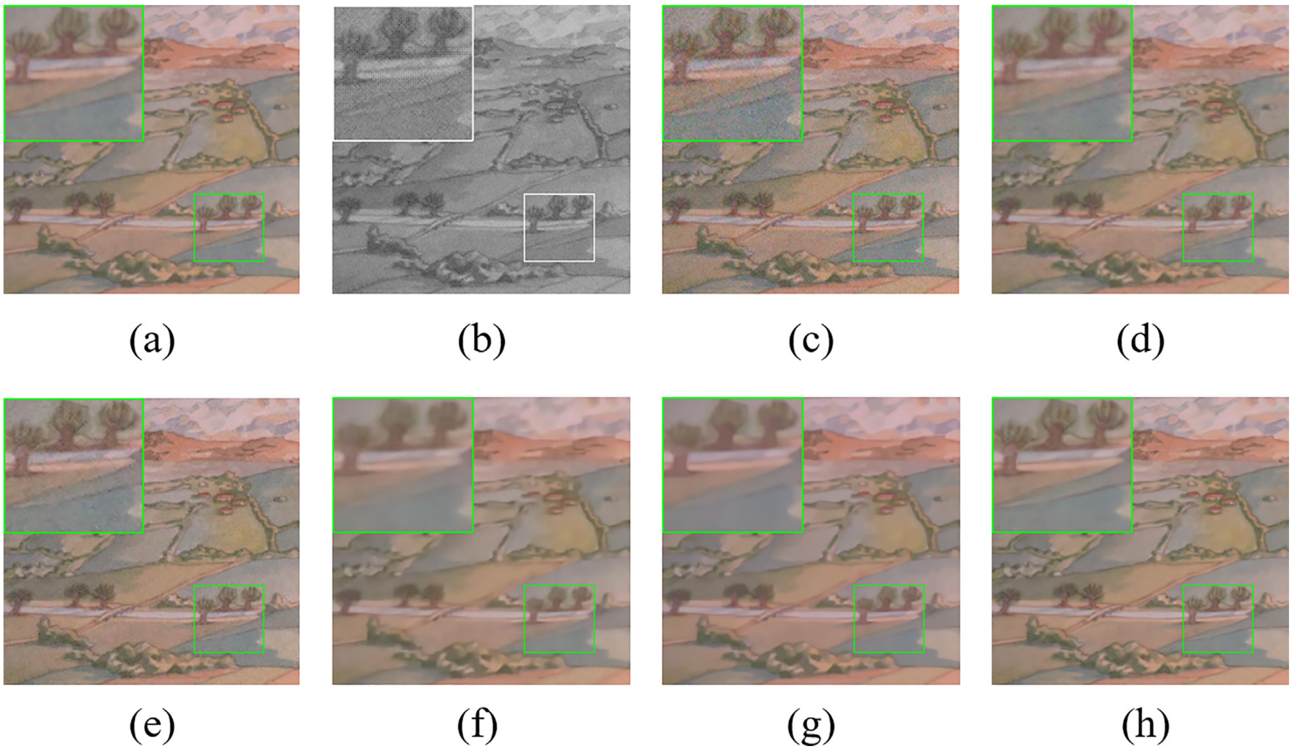

Fig. 11. Visual comparison results for the real noisy images from the SIDD dataset. (a) The clean image; (b) the noisy raw image; (c) the noisy sRGB image (PSNR = 30.17 dB, SSIM = 0.6713); (d) FlexISP [38] (PSNR = 33.36 dB, SSIM = 0.8200); (e) SEM [40] (PSNR = 34.37 dB, SSIM = 0.8514); (r) DJ [22] (PSNR = 35.40 dB, SSIM = 0.9163); (g) ADMM [59] (PSNR = 36.82 dB, SSIM = 0.9437); (h) **Proposed method** (PSNR = **38.40** dB, SSIM = **0.9496**).

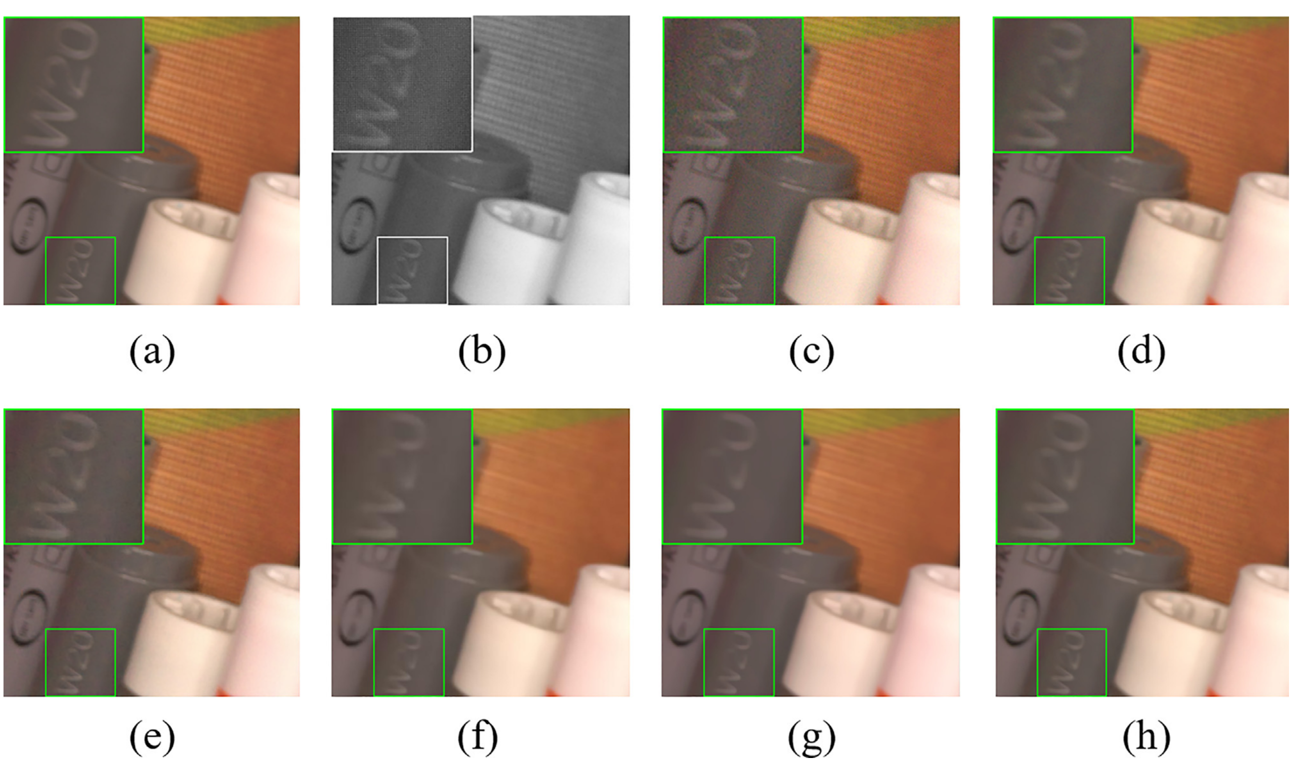

Fig. 12. Visual comparison results for the real noisy images from the SIDD dataset. (a) The clean image; (b) the noisy raw image; (c) the noisy sRGB image (PSNR = 32.63 dB, SSIM = 0.7431); (d) FlexISP [38] (PSNR = 37.35 dB, SSIM = 0.8932); (e) SEM [40] (PSNR = 39.23 dB, SSIM = 0.9404); (r) DJ [22] (PSNR = 39.36 dB, SSIM = 0.9582); (g) ADMM [59] (PSNR = 39.19 dB, SSIM = 0.9696); (h) **Proposed method** (PSNR = **42.27** dB, SSIM = **0.9726**).

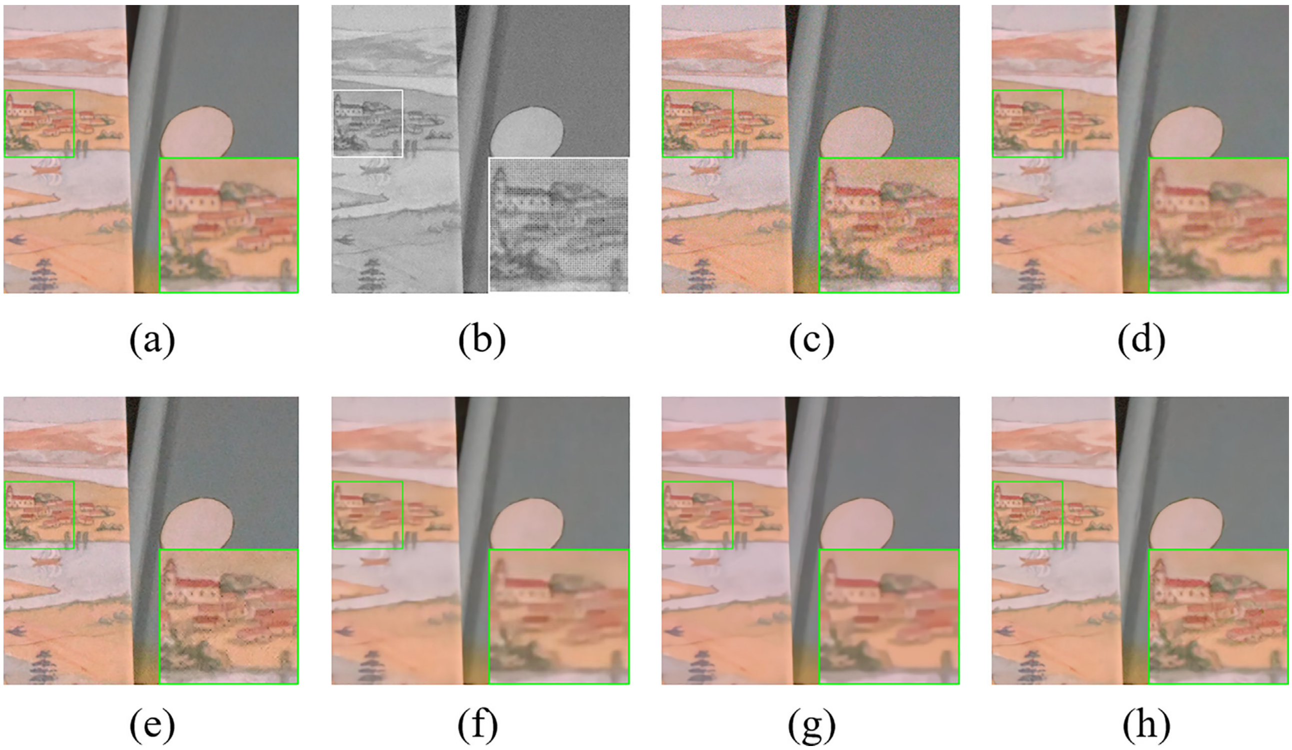

Fig. 13. Visual comparison results for the real noisy images from the SIDD dataset. (a) The clean image; (b) the noisy raw image; (c) the noisy sRGB image (PSNR = 30.19 dB, SSIM = 0.6467); (d) FlexISP [38] (PSNR = 33.84 dB, SSIM = 0.8103); (e) SEM [40] (PSNR = 34.45 dB, SSIM = 0.8384); (f) DJ [22] (PSNR = 36.55 dB, SSIM = 0.9171); (g) ADMM [59] (PSNR = 36.72 dB, SSIM = 0.9313); (h) **Proposed method** (PSNR = **38.25** dB, SSIM = **0.9352**).

method in this comparison study group, by up to $4.38dB, 4.64dB, 3.10dB$ and $2.87dB$ on the four datasets. Parts of the recovered color images by the test methods are shown in Figs. 6 and 7, from which one can observe that the proposed method can recover more details of the textures and fine structures in reconstructed images.

We have also verified the performance of the proposed JDD method on the MSR dataset [39], which contains 200 noise-free and noisy raw image pairs, and the corresponding RGB images. The noisy raw CFA images were synthetically generated by the Poisson-Gaussian noise model, where the parameters of the noise model are unknown. As the proposed method needs the noise variance maps as labels to train the noise estimation subnetwork, we simply estimate the noise levels σ_n in a local window of size 5×5 for each pixel using the pairs of noisy raw and noise-free raw images. Similar to other competing methods, we first pretrain the proposed method on the training dataset described in Section 3.4 and then fine tune the proposed method on the training images of the MSR dataset [39] for both noise-free and noisy cases. Table 3 shows the average PSNR results of the competing methods (i.e., the LNRF [39], SEM [40], DJ [22], CCRDN [24] and the MMNet [25] methods) on the MSR dataset. From Table 3 one can see that the proposed method performs comparably with the MMNet method of [25] for the noise free case. For the noise free case, the DJ method [22] performs best, as it adopted a specific training strategy for the hard samples of the dataset for better performance. For the noisy case, the proposed method outperforms all the other competing methods, verifying the superiority of the proposed blind joint demosaicing and denoising method. Parts of the reconstructed images by the test methods are shown in Figs. 8 and 9, from which we can see that our method can reproduce more visually pleasant images than the other competing methods.

4.3. Experimental results on real noisy images

Finally, we have evaluated the performance of the proposed method on a set of real noisy CFA images of the Smartphone Image Denoising Dataset (SIDD) [44]. We randomly select 30 images from the normal brightness category of the SIDD, and then randomly crop 300 patches of size 512×512 from these images without overlapping. We evaluate the performance of the competing methods on the 300 real noisy RAW image patches. For fairness, we didn't fine tune the proposed method on the SIDD. Table 4 shows the average PSNR/SSIM results on the set of real noisy CFA images, from which we can see that the proposed method performs significantly better than other competing methods. Figs. 10–13 show the visual comparison with other competing methods. From these images, one can see that our method can more effectively suppress the noise and recover more details around the edge and texture regions.

5. Conclusions

This paper presented a powerful blind joint demosaicing and denoise scheme based on noise level estimation and GAN. We have developed an end-to-end optimization technique using a combination of perceptual and adversarial loss functions. The introduction of discriminator network and end-to-end optimization makes it possible to achieve the quality assurance in the challenging scenario of blind JDD even in the presence of signal-dependent noise variations. The proposed GAN-based approach not only significantly improves the visual quality of reconstructed images but also keep the computational cost comparable to that of other competing approaches. Experimental results on real noisy raw images demonstrate that the effectiveness of the proposed method for the blind joint demosaicing and denoising.

CRediT authorship contribution statement

Fangfang Wu: Conceptualization, Methodology, Software. **Tao Huang:** Software, Validation. **Weisheng Dong:** Conceptualization, Writing - original draft. **Guangming Shi:** Supervision, Resources. **Zhonglong Zheng:** Investigation, Validation. **Xin Li:** Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Fangfang Wu received the B.S. degree in electronic engineering from the Xidian University, Xi'an, China, in 2008. She is currently pursuing the Ph.D. degree in Intelligent Information Processing at Xidian University, Xi'an, China. Her research interests include compressive sensing, Sparse representation, and deep learning.

Tao Huang received the B.S. degree in electronic engineering from Xidian University, Xi'an, China, in 2018. He is currently pursuing the Ph.D. degree in Electronic Science and Technology at Xidian University, Xi'an, China. His research interests include image restoration and deep learning.

Weisheng Dong received the B.S. degree in electronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010. He was a Visiting Student with Microsoft Research Asia, Bejing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic Engineering, Xidian University, as a Lecturer, where he has been a Professor since 2016. His research interests include inverse problems in image processing, deep learning, parse representation. He was a recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing (VCIP) in 2010. He has served as an associate editor of IEEE Transactions on Image Processing and is currently an associate editor of SIAM Journal of Imaging Sciences.

Guangming Shi received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph. D. degree in electronic information technology from Xidian University, Xi'an, China, in 1985, 1988, and 2002, respectively. He had studied at the University of Illinois and University of Hong Kong. Since 2003, he has been a Professor with the School of Electronic Engineering, Xidian University. He awarded Cheung Kong scholar Chair Professor by ministry of education in 2012. He is currently the Academic Leader on circuits and systems, Xidian University. His research interests include compressed sensing, brain cognition theory, multirate filter banks, image denoising, low-bitrate image and video coding, and implementation of algorithms for intelligent signal processing. He has authored or co-authored over 200 papers in journals and conferences. He served as the Chair for the 90th MPEG and 50th JPEG of the international standards organization (ISO), technical program chair for FSKD06, VSPC 2009, IEEE PCM 2009, SPIE VCIP 2010, IEEE ISCAS 2013.

Zhonglong Zheng received the B.S. degree in electronic engineering from the Petroleum University, China, in1999, and the Ph.D. degree in pattern recognition and intelligent system from Shanghai Jiaotong University, Shanghai, China, in 2005. In 2006, he joined the Zhejiang Normal University, as a Lecturer. Since 2011, he has been a Professor in the department of CS. From 2012 to 2013, he was a visiting scholar with the EECS, University of California at Merced, USA. His research interests include machine learning and computer vision.

Xin Li received the B.S. (Hons.) degree in electronic engineering and information science from the University of Science and Technology of China, Hefei, China, in 1996, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 2000. He was a Technical Staff Member with the Sharp Laboratories of America, Camas, WA, USA, from 2000 to 2002. Since 2003, he has been a Faculty Member with the Lane Department of Computer Science and Electrical Engineering. His research interests include image/video coding and processing. He was a recipient of the Best Student Paper Award at the Conference of Visual Communications and Image Processing in 2001, the Best Student Paper Award at the IEEE Asilomar Conference on Signals, Systems and Computers in 2006, and the Best Paper Award at the Conference of Visual Communications and Image Processing in 2010. He is currently serving as a member of the Image, Video and Multidimensional Signal Processing Technical Committee and an Associate Editor of the IEEE Transactions on Circuits and Systems for Video Technology.